NATIONAL UNIVERSITY OF SINGAPORE
Department of Statistics and Data Science

ST5202: Applied Regression Analysis

TUTORIAL 1 (Solution)

1. The data presented in the Table below was published in the March 1, 1984 issue of the Wall Street Journal. It relates to the advertising budget (in millions of dollars) of 21 firms for 1983 and millions of impressions retained per week by the viewers of the products of these firms. The data are based on a survey of 4000 adults in which users of the products were asked to cite a commercial they had seen for the product category in the past week.

IMPACT OF ADVERTISING EXPENDITURE

Firm	Impressions, millions	Expenditure, millions of 1983 dollars
1. Miller Lite	32.1	50.1
2. Pepsi	99.6	74.1
3. Stroh's	11.7	19.3
4. Fed'l Express	21.9	22.9
5. Burger King	60.8	82.4
6. Coca Cola	78.6	40.1
7. McDonald's	92.4	185.9
8. MCI	50.7	26.9
9. Diet Cola	21.4	20.4
10. Ford	40.1	166.2
11. Levi's	40.8	27.0
12. Bud Lite	10.4	45.6
13. ATT/Bell	88.9	154.9
14. Calvin Klein	12.0	5.0
15. Wendy's	29.2	49.7
16. Polaroid	38.0	26.9
17. Shasta	10.0	5.7
18. Meow Mix	12.3	7.6
19. Oscar Meyer	23.4	9.2
20. Crest	71.1	32.4
21. Kibbles 'N Bits	4.4	6.1

Source: http://lib.stat.cmu.edu/DASL/Datafiles/tvadsdat.html

(i) Create a `.txt` for the data given in the above table so that you can input into you R session.

```
    impre adver
1    32.1  50.1
2    99.6  74.1
3    11.7  19.3
4    21.9  22.9
5    60.8  82.4
6    78.6  40.1
7    92.4 185.9
8    50.7  26.9
9    21.4  20.4
10   40.1 166.2
11   40.8  27.0
12   10.4  45.6
13   88.9 154.9
14   12.0   5.0
15   29.2  49.7
16   38.0  26.9
17   10.0   5.7
18   12.3   7.6
19   23.4   9.2
20   71.1  32.4
21    4.4   6.1
```

The `.txt` *file is prepared as above.*

Note: The first column for index is optional. For the other two columns, each should have a heading specifying the variable name. The created data file is named `AdverImpact.txt`*.*

(ii) Write R codes to do the following:

(iia) Input the data from the data file you created into R.

(iib) Make the variables in the data frame available in your R session.

(iic) Plot the response variable against the predictor variable.

(iid) Fit a simple regression model to the data and extract the fitted results.

(iiie) Plot the residuals against the fitted values.

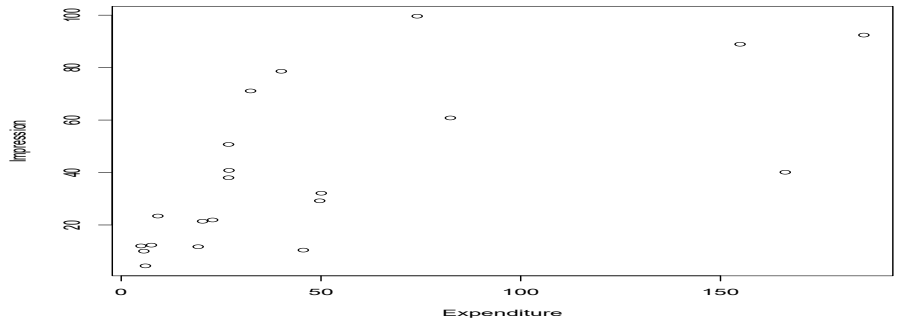
The R codes are given below:

```
q1.dat=read.table("D://Rsession/AdverImpact.txt",header=TRUE) # (iia)
y=q1.dat$impre; x=q1.dat$adver                                 # (iib)
plot(x, y, xlab='Expenditure', ylab='Impression')              # (iic)
q1.fit = lm(y~x); summary(q1.fit)                              # (iid)
r = q1.fit$resid
fitted = q1.fit$fitted
plot(fitted, r, xlab='Fitted values', ylab='Residuals')        # (iie)
```

(iii) From the results obtained by implementing your R codes, answer the following questions:

(iiia) What can you say about the nature of the relationship between the two variables from the plot in (iic)?

The plot in (iic) is given below:



There is a linear trend in the scatter plot, which suggests that a linear relationship exists between the two variables.

(iiib) What is the estimated regression function?

The R codes in (iiid) produce the following results:

```
Estimate Std. Error t value Pr(>|t|)
(Intercept) 22.16269    7.08948   3.126  0.00556
x            0.36317    0.09712   3.739  0.00139
```

From the results, we obtain the fitted regression function as

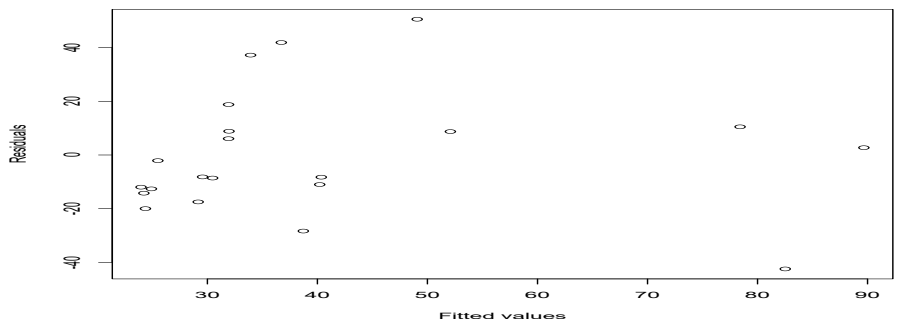
$$\hat{y} = 22.1627 + 0.3632x.$$

(iiic) If a firm decides to increase its current advertisement budget by two millions of dollars, what will be the expected increase in the impressions (in millions) on its products? Provide an estimate of the expected increase.

From the regression function, when x increases to $x+2$, the expected impressions increase from $\beta_0 + \beta_1 x$ to $\beta_0 + \beta_1(x+2)$. Hence, the expected increase is $2\beta_1$. Its estimate is given by $2\hat{\beta}_1 = 2 * 0.3532 = 0.7064$.

(iiid) From the residual plot, can you find any unusual points?

The following is the residual plot:



In the residual plot, the three right most points are further away from the majority. In addition, the plot does not look like a random pattern (the random pattern is that points are scattered evenly within a horizontal band). Since the range along the Y-axis gets larger as the fitted value gets larger, it suggests that the variance might not be a constant.

2. For 120 graduate students, their ACT and GPA are recorded. The data is given in the file `GPA.txt` on Canvas. Suppose that the purpose of the study is to establish a model for the prediction of GPA by ACT.

   (i) Determine which variable is to be taken as the response variable. Draw the scatter plot of the response variable against the predictor variable, impose the estimated regression line on the plot. Does the estimated regression appear to fit the data well?

   GPA is the response variable, ACT is the predictor variable. The following linear regression model can be explored:

$$GPA = \beta_0 + \beta_1 ACT + \epsilon.$$

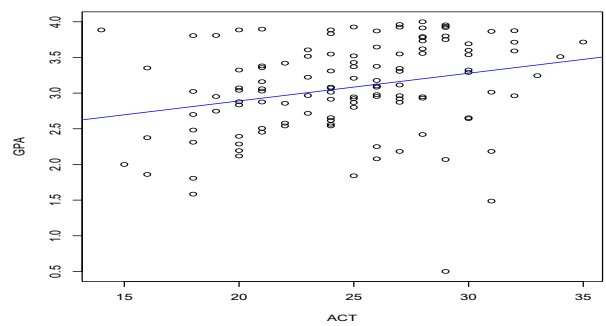
   The model is fitted by the following codes:

```
q2.dat = read.table("D://Rsession/GPA.txt"); attach(q2.dat)
q2.fit = lm(GPA~ACT,data=q2.dat); summary(q2.fit)
```

   The plot is obtained by

```
plot(ACT,GPA);  abline(q2.fit,col="blue")
```

The plot is as follows:



The regression model does not seem to fit well. The scatter plot has a curvature pattern.

(ii) How much of the variation in the response variable is explained by the predictor variable?

The computed results are as follows.

```
Estimate Std. Error t value Pr(>|t|)
(Intercept)  2.11405    0.32089   6.588  1.3e-09 ***
ACT          0.03883    0.01277   3.040  0.00292 **
Residual standard error: 0.6231 on 118 degrees of freedom
Multiple R-squared:  0.07262,   Adjusted R-squared:  0.06476
F-statistic:  9.24 on 1 and 118 DF,  p-value: 0.002917
```

From the summary results, $R^2 = 0.07262$*, about 7% of the variation in GPA is explained by ACT.*

(iii) If a student has an ACT score three unites higher than another student, will the first student have a higher expected GPA than the second student? What is the estimated expected GPA difference between the two students?

From the regression model, the difference between the first and the second student in expected GPA is $3\beta_1$*. Whether the first one has higher expected GPA depends*

on whether $\beta_1 > 0$. The estimate of the difference is $3\hat{\beta}_1 = 3 * 0.03883 = 0.11649$. Hence,the first one has higher expected GPA.

(iv) A student has an ACT score 30. Obtain a point prediction of his GPA both by manual computation using the prediction formula and by using the R function `predict`.

Manually, we have

$$\widehat{GPA} = 2.11405 + 0.03883 \times 30 = 3.27895.$$

From `predict(q2.fit, list(ACT=30))`*, we get $GPA = 3.278863$. The difference is due to rounding errors.*

3. A criminologist studying the relationship between population density and robbery rate in medium-sized U.S. cities collected the following data for a random sample of 16 cities; X is the population density of the city (number of people per unit area), and Y is the robbery rate last year (number of robberies per 100,000 people).

i	1	2	$\cdots$	16
X_i	59	49	$\cdots$	70
Y_i	209	180	$\cdots$	204

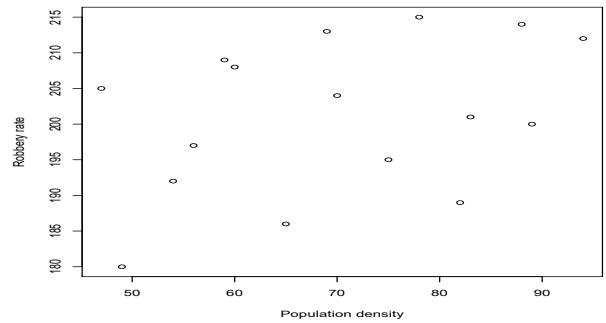
The full data is provided in the file `robbery.txt` on Canvas.

(i) Draw a scatter plot of Y against X. What relationship between Y and X does the plot suggest?

The plot is obtained by the following R-codes:

```
q3.dat = read.table("D://Rsession/robbery.txt")
Y = q3.dat$V1; X = q3.dat$V2
plot(X,Y,xlab="Population density",ylab="Robbery rate")content...
```

The plot is as follows:



The plot suggests that there is a linear relationship between the two variables.

(ii) Fit a simple linear regression model to the data and obtain the estimated regression function. Impose the estimated regression line on the scatter plot obtained in (i). Does the linear regression function appear to give a good fit? Discuss.

The additional R codes for this part is as follows:

```
q3.fit = lm(Y~X); summary(q3.fit)
plot(X,Y,xlab="Population density",ylab="Robbery rate")
abline(q3.fit,col="blue")
```

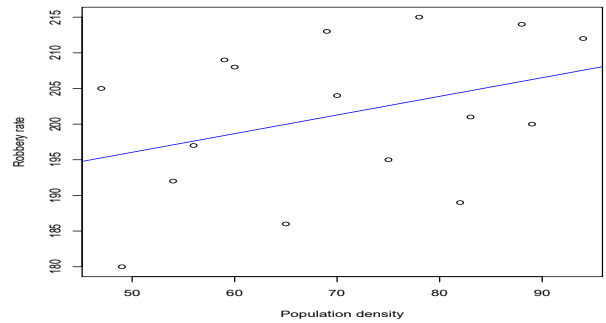
The fitted results are:

```
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 182.9725    12.7223  14.382 8.87e-10 ***
X             0.2616     0.1783   1.467    0.164
Residual standard error: 10.29 on 14 degrees of freedom
Multiple R-squared:  0.1332,    Adjusted R-squared:  0.07132
F-statistic: 2.152 on 1 and 14 DF,  p-value: 0.1645
```

The estimated regression function is

$$\hat{Y} = 182.9725 + 0.2616X.$$

The plot is as follows:



The plot shows that the linear relationship is appropriate.

(iii) Obtain point estimates of the following: (1) the difference in the expected robbery rate in cites that differ by one unit in population density, (2) the expected robbery rate last year in cities with population density $X = 60$, (3) ϵ_{10}, (4) σ^2.

(1) The difference is β_1, the estimate is 0.2616.

(2) The estimate is given by $\hat{\beta}_0 + \hat{\beta}_1 \times 60 = 182.9725 + 0.2616 \times 60 = 198.6685$.

(3) ϵ_{10} is estimated by e_{10}, the residual. The value is extracted by `r = q3.fit$resid;` $e_{10} = r[10] = -3.683162$.

(4) From the fitted results, $\hat{\sigma} = 10.29$. Hence $\hat{\sigma}^2 = 10.29^2 = 105.8841$.

4. Experience with a certain type of plastic indicates that a relation exists between the hardness of items molded from the plastic (y, in Brinell units) and the elapsed time since termination of the molding process (x, in hours). Twelve batches of the plastic were made, and from each batch one test item was molded. The hardness and the elapsed time are measured for each item. The data is given below:

x	32	48	72	64	48	16	40	48	48	24	80	56
y	230	262	323	298	255	199	248	279	267	214	359	305

(i) It is proposed to study this relation by means of regression analysis. A participant in the discussion objects, pointing out that "the hardening of the plastic is the result of a natural chemical process that doesn't leave anything to chance, so the relation

must be mathematical and regression analysis is not appropriate." Is the objection reasonable? Explain.

No. In addition to elapsed time, there are still other uncertain factors which could affect the hardness such as room temperature, humility, etc. Given the same elapsed time, the hardness can still vary. It is evidenced by the data: there are four items whose elapsed time are all 48, but their hardness are all different.

(ii) Obtain the estimated regression function. Plot the estimated regression function and the data. Does a linear regression function appear to give a good fit here?

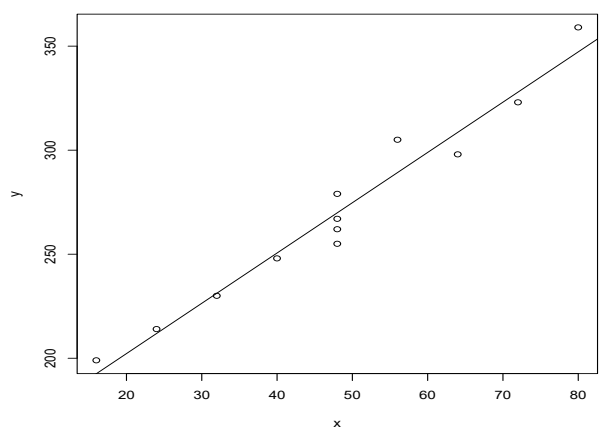
The model fitting and plots are carried out by the following codes

```
y=q4.dat$y; x=q4.dat$x;
fit=lm(y~x); summary(fit)
plot(x,y); abline(fit)
```

The summary results are:

```
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 153.9167     8.0666   19.08 3.40e-09
x             2.4167     0.1575   15.35 2.81e-08
```

The plot is as follows.



From the fitted result, the fitted regression function is obtained as

$$\hat{Y} = 153.9167 + 2.4167X.$$

In the plot, the data points are scattered quit evenly along the fitted regression line. It indicates that the regression function does give a good fit.

(iii) Obtain a point estimate of the mean hardness when $x = 48$ hours.

The estimate is given by

$$153.9167 + 2.4167 * 48 = 269.9183.$$

(iv) Obtain a point estimate of the change in mean hardness when x increases by one hour.

The change is

$$\beta_0 + \beta_1(x + 1) - (\beta_0 + \beta_1 x) = \beta_1.$$

The estimate is $\hat{\beta}_1 = 2.4167$.